



Lab Worksheet 2

CSE461: Introduction to Robotics

Department of Computer Science and Engineering

Lab 02: Introduction to Raspberry PI and control an LED with a switch using Raspberry PI.

I. Topic Overview

The lab is designed to introduce the basics of a microcontroller using Raspberry PI development board along with Raspberry PI OS. The students will get introduced to the pins of the board, their functions and lastly, learn how to use the GPIO - General Purpose Input Output pins for interfacing simple I/O devices like LED and switches with the board. The aim of this lab is to gain a foundational understanding of Raspberry PI and use it in circuit design.

II. Learning Outcome

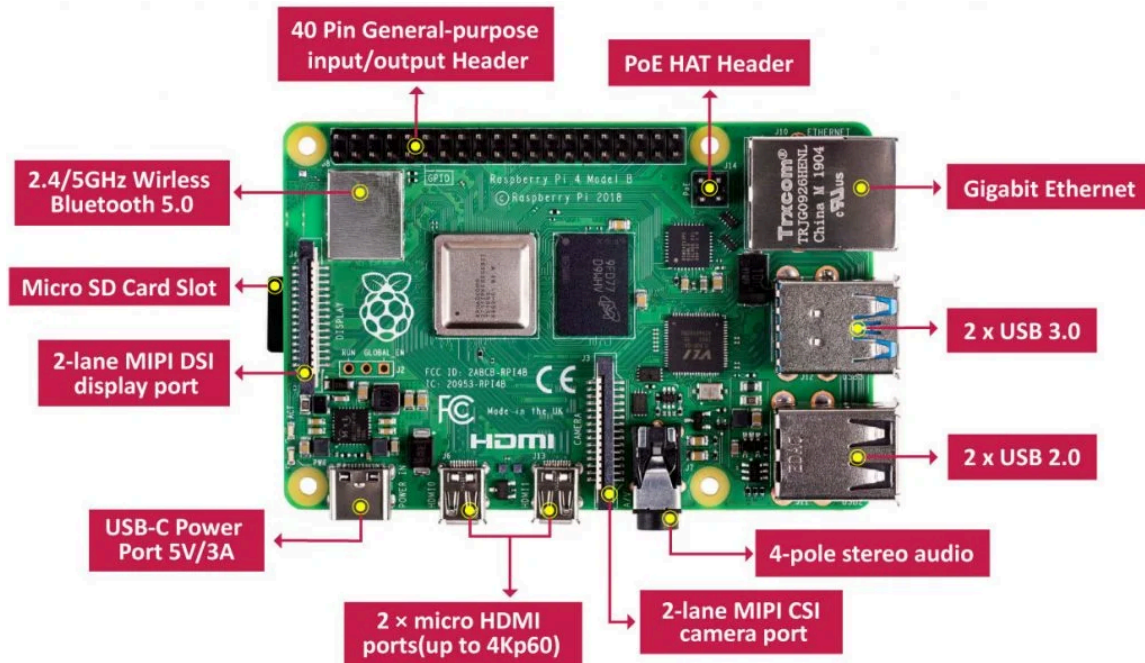
After this lab, students will be able to:

1. Know the various pins of the Raspberry PI board and their functions.
2. Build and troubleshoot simple circuits using LEDs, resistors, and switches..
3. Code to configure the GPIO pins to control different circuits for different usages.
4. Learn how to install Python libraries on the Raspberry Pi and how to create and run Python programs using the terminal.

III. Materials

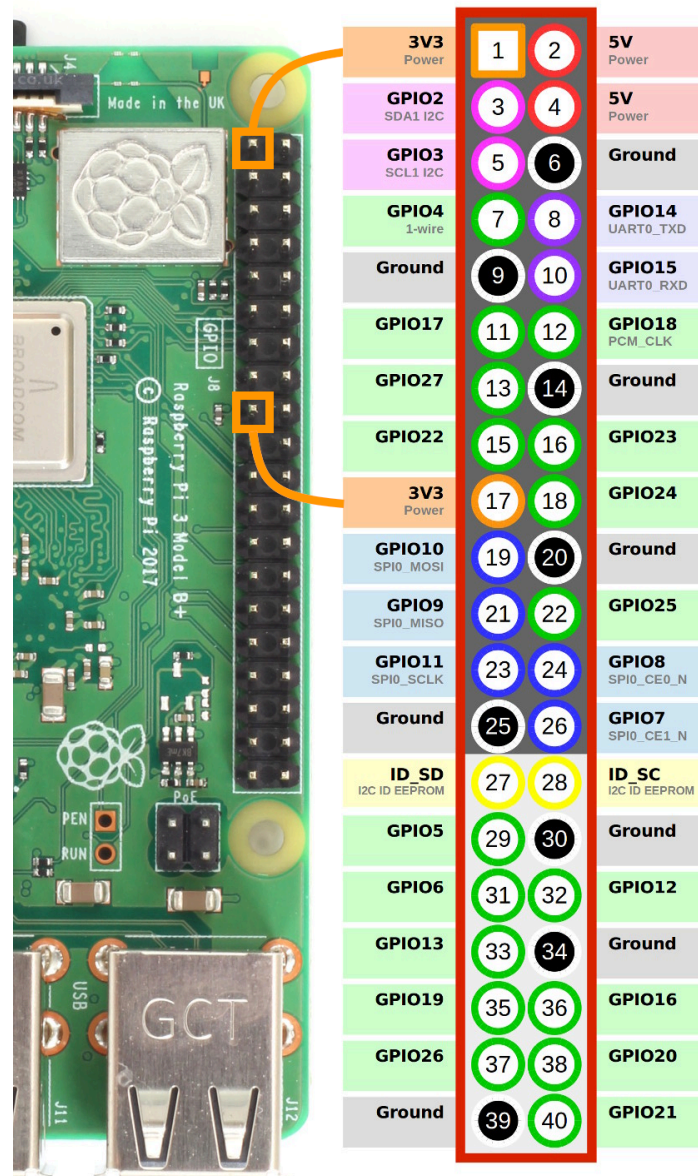
- Raspberry PI development board
- LED (Light Emitting Diode)
- Resistor (appropriate value for current limiting, typically around 220 ohms)
- Switch
- USB pen drive containing OS

IV. Raspberry PI Overview



Raspberry Pi 4 Model B features a high-performance 64-bit quad-core processor, dual-display support at resolutions up to 4K via a pair of micro HDMI ports, hardware video decode at up to 4Kp60, up to 8GB of RAM, dual-band 2.4/5.0 GHz wireless LAN, Bluetooth 5.0, Gigabit Ethernet, USB 3.0, and PoE capability (via a separate PoE HAT add-on). For the end user, the Raspberry Pi 4 Model B provides desktop performance comparable to entry-level x86 PC systems. Raspberry Pi OS (previously called Raspbian) is the recommended operating system for normal use on a Raspberry Pi. [Raspberry Pi Imager](#) is the quick and easy way to install Raspberry Pi OS and other operating systems to a microSD card or a pendrive, ready to use with your Raspberry Pi. Then the board can be programmed using user-friendly IDEs; for example, Python programming can be done using Thonny IDE.

V. Raspberry Pi Pins:



The Raspberry PI 4B has the following types of pins:

1. Power Pins:

- 5V and 3.3V:** Provide regulated voltage output.
- GND:** Ground pins.

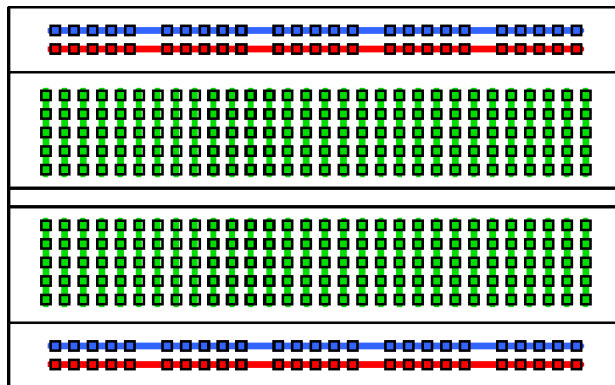
2. **GPIO Pins:** The general purpose input and output pins. Some of these GPIO pins can be used for following different communication protocols like UART, I2C and SPI to communicate with other devices. The GPIO pins work at a high voltage of 1.8V-3.3V maximum.

V. Raspberry PI OS Installation Guideline: Setting up Raspberry PI

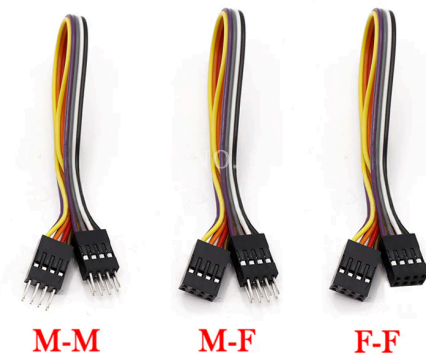
VII. Experiment 1: LED blinking.

1. Description the components:

a. Breadboard



b. Jumper wires

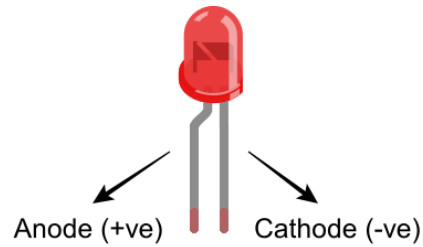


c. Resistors

4 Band Resistor Color Code				
Color	1. Band	2. Band	3. Band Multiplier	4. Band Tolerance
black	-	0	1	-
brown	1	1	10	+/- 1%
red	2	2	100	+/- 2%
orange	3	3	1'000	-
yellow	4	4	10'000	-
green	5	5	100'000	-
blue	6	6	1'000'000	-
purple	7	7	-	-
grey	8	8	-	-
white	9	9	-	-
gold	-	-	0.1	+/- 5%
silver	-	-	0.01	+/- 10%

Table 1: 4 band resistor color codes

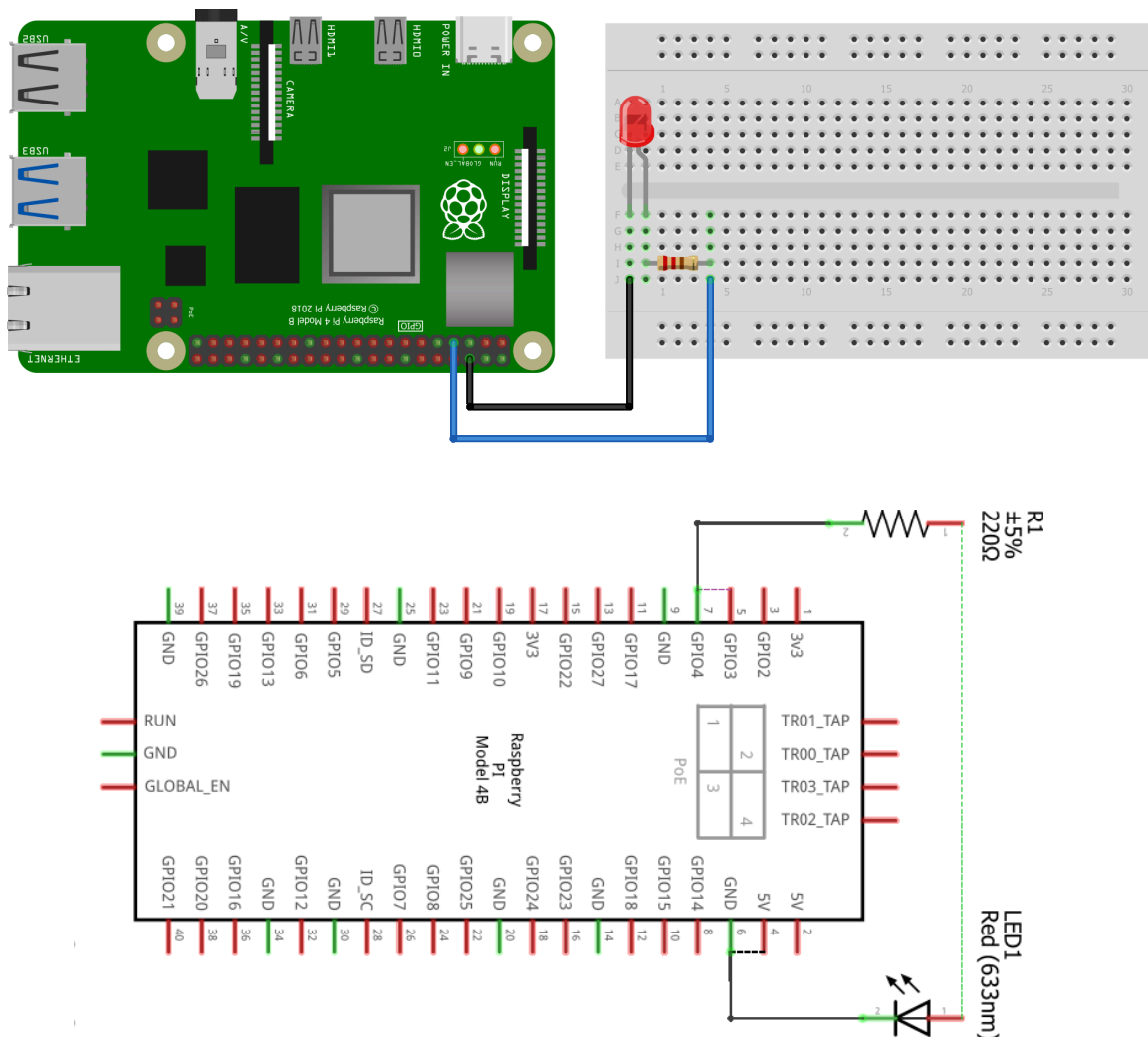
d. LED



2. Setting up the circuit:

- Connect one end of a 220 ohm resistor to a GPIO pin of your Raspberry PI and connect the other end to the anode of the LED.
- Connect the cathode of the LED to a ground pin of the board.

3. Circuit diagram:



4. **Code:** Open a .py file in a folder on the RPI OS. Open Thonny IDE and save the code in the previously created file. Then run the code.

```
from gpiozero import LED
#imports LED functions from gpiozero library
import time
#imports time library

led = LED(4)
#declare the GPIO pin 4 for LED output and store it in led variable

while True:
#initiated an infinite while loop
    led.on()
    #turn on the led
    time.sleep(1)
    #delay for 1 second (increase the delay if necessary)
    led.off()
    #turn off the led
    time.sleep(1)
    #delay for 1 second (increase the delay if necessary)
```

VIII. Experiment 2: Printing “Switch pressed.” on the terminal using a switch.

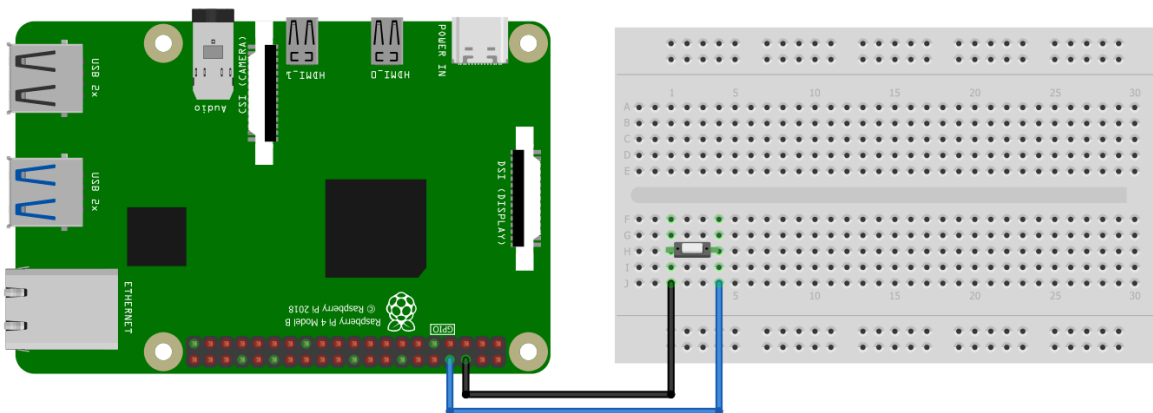
1. **Description the components:**

- Push-button switch

2. **Setting up the circuit:**

- a. Connect one terminal of the switch to the GND pin.
- b. Connect the other terminal of the switch to a GPIO pin (GPIO 14)

3. **Circuit diagram:**



```
from gpiozero import Button
#imports Button functions from gpiozero library
import time
#imports time library

button = Button(14)
#declare the GPIO pin 14 for button input and store it in button
variable

while True:
#initiated an infinite while loop
    if button.is_pressed:
        #checks if button is pressed
        print("Switch pressed")
    else:
        #if button is not pressed
        print("Not pressed")
    time.sleep(0.5)
    #delay for half a second
```

VIII. Lab Task: Control LED using a switch.

Task Description:

Modify the circuit and code from Experiments 1 and 2 to turn ON an LED only when the switch is pressed. The LED should turn OFF when the switch is released.

Deliverables:

- Circuit diagram
- Raspberry PI code
- Demonstration of the working circuit

IX. References:

1. <https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-datasheet.pdf>
2. <https://www.raspberrypi.com/documentation/computers/getting-started.html>